# Chapter 3

## The Relational Data Model
## and
## Relational Database Constraints

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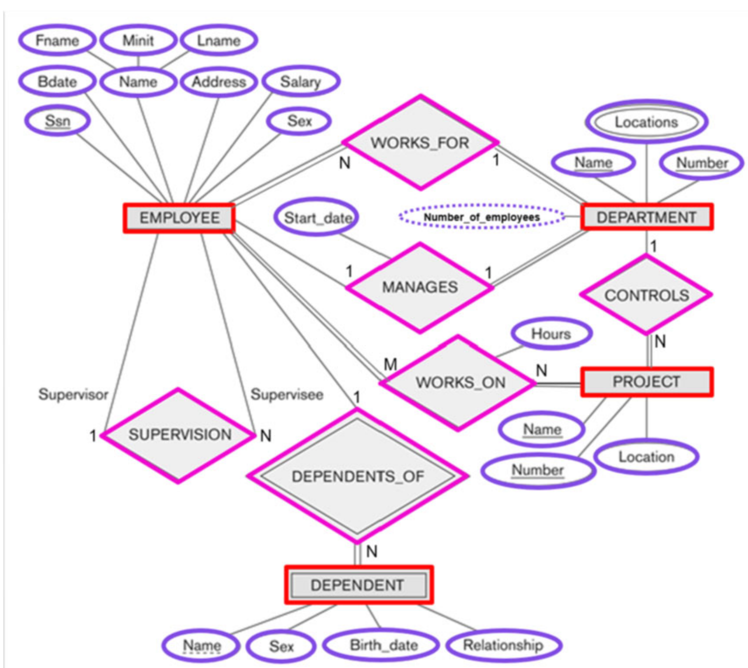


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# Informal Definition

Informally, a **relation** looks like a **table** of values.

Relation Name

Attributes

**STUDENT**

| Name | Ssn | Home_phone | Address | Office_phone | Age | Gpa |
|---|---|---|---|---|---|---|
| Benjamin Bayer | 305-61-2435 | 373-1616 | 2918 Bluebonnet Lane | NULL | 19 | 3.21 |
| Chung-cha Kim | 381-62-1245 | 375-4409 | 125 Kirby Road | NULL | 18 | 2.89 |
| Dick Davidson | 422-11-2320 | NULL | 3452 Elgin Road | 749-1253 | 25 | 3.53 |
| Rohan Panchal | 489-22-1100 | 376-9821 | 265 Lark Lane | 749-6492 | 28 | 3.93 |
| Barbara Benson | 533-69-1238 | 839-8461 | 7384 Fontana Lane | NULL | 19 | 3.25 |

Tuples

key

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# Formal Definitions - Schema

The **Schema** (or description) of a Relation:

- Denoted by $R(A_1, . A_2, .., A_n)$
- $R$ is the **name** of the relation (table)
- The **attributes** of the relation are $A_1, . A_2, .., A_n$

**Example:**

CUSTOMER (CustId, Cust-name, Address, PhoneNo)

relation name

attributes

6 digit numbers

Each attribute has a **domain** or a set of valid values.

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## Formal Definitions - Tuple

A **tuple** (row) is an ordered set of values (in angled brackets <...>)

Each value is taken from the appropriate **domain**.

A row in the CUSTOMER relation is a 4-tuple consisting of four values:

< 632895, "John Smith", "101 Main St., Atlanta, GA 30332", "404–894–2000" >

A **relation** (table) is a **set** of such tuples (rows)

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## The Relation STUDENT (schema and state)

Relation Name

Attributes

**STUDENT**

| Name | Ssn | Home_phone | Address | Office_phone | Age | Gpa |
|---|---|---|---|---|---|---|
| Benjamin Bayer | 305-61-2435 | 373-1616 | 2918 Bluebonnet Lane | NULL | 19 | 3.21 |
| Chung-cha Kim | 381-62-1245 | 375-4409 | 125 Kirby Road | NULL | 18 | 2.89 |
| Dick Davidson | 422-11-2320 | NULL | 3452 Elgin Road | 749-1253 | 25 | 3.53 |
| Rohan Panchal | 489-22-1100 | 376-9821 | 265 Lark Lane | 749-6492 | 28 | 3.93 |
| Barbara Benson | 533-69-1238 | 839-8461 | 7384 Fontana Lane | NULL | 19 | 3.25 |

Tuples

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## Characteristics of Relations

The tuples (rows) in a relation (table) are a set, thus **not ordered**, even though they appear to be in the table.

The attributes (columns) in a tuple **are ordered**.

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## Relational Integrity Constraints

Constraints are **conditions** that must hold on **all** valid relation states:

- **Domain** constraints
- **Key** constraints
- **Entity integrity** constraints
- **Referential integrity** constraints

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## Domain Constraints

Every value in a tuple must be from the **domain of its attribute**
(or it could be **null**, if allowed for that attribute)

A valid row in the CUSTOMER relation:

< **632895**, "John Smith", "101 Main St., Atlanta, GA 30332", "404–894–2000" >

An invalid row in the CUSTOMER relation:

< **A65F76**, "John Smith", "101 Main St., Atlanta, GA 30332", "404–894–2000" >



## Key Constraints

A **superkey** of $R$ is a set of attributes $SK$ of $R$ such that no two tuples in any valid state of $R$ have the same values for $SK$

A **key** of $R$ *is a* "minimal" superkey, i.e. removal of any attribute from $K$ results in a set of attributes that is not a superkey.

No two tuples of $R$ may have the same key value.

**Example**: In the CAR relation schema:

CAR (State, RegNo, SerialNo, Make, Model, Year)

CAR has two keys:

- Key1 = {State, RegNo}
- Key2 = {SerialNo}

Both are also superkeys of CAR

{SerialNo, Make} is a superkey but **not** a key.

For any two distinct tuples $t_1, t_2 \in r(R)$: $\quad t_1[K] \neq t_2[K]$

## Keys

| Student_ID | Student_Enroll | Student_Name | Student_Email |
|---|---|---|---|
| S02 | 4545 | Dave | dave@gmail.com |
| S34 | 4541 | Jack | jack@njit.edu |
| S22 | 4555 | Mark | mark1@gmail.com |
| S23 | 4545 | Mark | mark2@gmail.com |

Candidate keys:

{Student_ID}
{Student_Email}

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## Keys

If a relation has several **candidate keys,** one is chosen to be the **primary key.** The primary key attributes are **underlined.**

CAR

| <u>License_number</u> | Engine_serial_number | Make | Model | Year |
|---|---|---|---|---|
| Texas ABC-739 | A69352 | Ford | Mustang | 02 |
| Florida TVP-347 | B43696 | Oldsmobile | Cutlass | 05 |
| New York MPO-22 | X83554 | Oldsmobile | Delta | 01 |
| California 432-TFY | C43742 | Mercedes | 190-D | 99 |
| California RSK-629 | Y82935 | Toyota | Camry | 04 |
| Texas RSK-629 | U028365 | Jaguar | XJS | 04 |

<u>License_number</u> is the primary key
Engine_serial_number is a candidate key

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## Entity Integrity Constraint

The **primary key attributes** $PK$ of each relation schema $R$ cannot have null values in any tuple:

$$t[PK] \neq \textbf{null} \text{ for all tuples } t \in r(R)$$

This is because primary key values are used to **identify** the individual tuples.

If $PK$ has several attributes, **null** is not allowed in any of these attributes.

**Note:** Other attributes of $R$ may be constrained to disallow null values, even though they are not members of the primary key.

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## COMPANY Database Schema

EMPLOYEE

| Fname | Minit | Lname | Ssn | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|---|---|---|---|---|---|---|---|---|---|

DEPARTMENT

| Dname | Dnumber | Mgr_ssn | Mgr_start_date |
|---|---|---|---|

DEPT_LOCATIONS

| Dnumber | Dlocation |
|---|---|

PROJECT

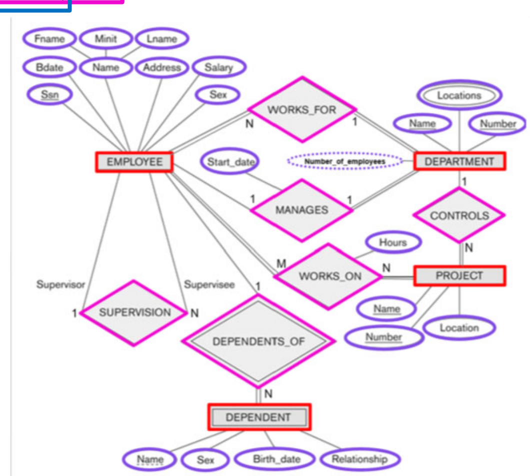
| Pname | Pnumber | Plocation | Dnum |
|---|---|---|---|

WORKS_ON

| Essn | Pno | Hours |
|---|---|---|

DEPENDENT

| Essn | Dependent_name | Sex | Bdate | Relationship |
|---|---|---|---|---|

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## Populated Database State for COMPANY

**EMPLOYEE**

| Fname | Minit | Lname | Ssn | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|---|---|---|---|---|---|---|---|---|---|
| John | B | Smith | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M | 30000 | 333445555 | 5 |
| Franklin | T | Wong | 333445555 | 1955-12-08 | 638 Voss, Houston, TX | M | 40000 | 888665555 | 5 |
| Alicia | J | Zelaya | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX | F | 25000 | 987654321 | 4 |
| Jennifer | S | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX | F | 43000 | 888665555 | 4 |
| Ramesh | K | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M | 38000 | 333445555 | 5 |
| Joyce | A | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX | F | 25000 | 333445555 | 5 |
| Ahmad | V | Jabbar | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX | M | 25000 | 987654321 | 4 |
| James | E | Borg | 888665555 | 1937-11-10 | 450 Stone, Houston, TX | M | 55000 | NULL | 1 |

**DEPT_LOCATIONS**

| Dnumber | Dlocation |
|---|---|
| 1 | Houston |
| 4 | Stafford |
| 5 | Bellaire |
| 5 | Sugarland |
| 5 | Houston |

**WORKS_ON**

| Essn | Pno | Hours |
|---|---|---|
| 123456789 | 1 | 32.5 |
| 123456789 | 2 | 7.5 |
| 666884444 | 3 | 40.0 |
| 453453453 | 1 | 20.0 |
| 453453453 | 2 | 20.0 |
| 333445555 | 2 | 10.0 |
| 333445555 | 3 | 10.0 |
| 333445555 | 10 | 10.0 |
| 333445555 | 20 | 10.0 |
| 999887777 | 30 | 30.0 |
| 999887777 | 10 | 10.0 |
| 987987987 | 10 | 35.0 |
| 987987987 | 30 | 5.0 |
| 987654321 | 30 | 20.0 |
| 987654321 | 20 | 15.0 |
| 888665555 | 20 | NULL |

**DEPARTMENT**

| Dname | Dnumber | Mgr_ssn | Mgr_start_date |
|---|---|---|---|
| Research | 5 | 333445555 | 1988-05-22 |
| Administration | 4 | 987654321 | 1995-01-01 |
| Headquarters | 1 | 888665555 | 1981-06-19 |

**PROJECT**

| Pname | Pnumber | Plocation | Dnum |
|---|---|---|---|
| ProductX | 1 | Bellaire | 5 |
| ProductY | 2 | Sugarland | 5 |
| ProductZ | 3 | Houston | 5 |
| Computerization | 10 | Stafford | 4 |
| Reorganization | 20 | Houston | 1 |
| Newbenefits | 30 | Stafford | 4 |

**DEPENDENT**

| Essn | Dependent_name | Sex | Bdate | Relationship |
|---|---|---|---|---|
| 333445555 | Alice | F | 1986-04-05 | Daughter |
| 333445555 | Theodore | M | 1983-10-25 | Son |
| 333445555 | Joy | F | 1958-05-03 | Spouse |
| 987654321 | Abner | M | 1942-02-28 | Spouse |
| 123456789 | Michael | M | 1988-01-04 | Son |
| 123456789 | Alice | F | 1988-12-30 | Daughter |
| 123456789 | Elizabeth | F | 1967-05-05 | Spouse |

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## Referential Integrity

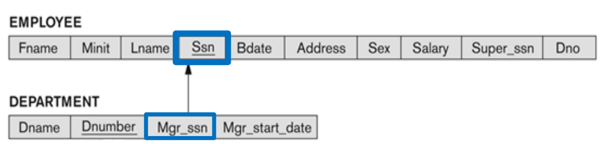
Used to specify a connection between tuples in *two* relations:

- The **referencing relation** and the **referenced relation.**

Tuples in the **referencing relation** $R_1$ have **foreign key** attributes $FK$ that reference the primary key attributes $PK$ of the **referenced relation** $R_2$.

- A tuple $t_1 \in R_1$ is said to **reference** a tuple $t_2 \in R_2$ if $t_1[FK] = t_2[PK]$

A referential integrity constraint is displayed in a relational database schema as an arrow from $R_1.FK$ to $R_2.PK$.

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## Referential Integrity (or Foreign Key) Constraint

The value in the foreign key column (or columns) $FK$ of the **referencing relation** $R_1$ can be either:

- a value of an existing primary key value of a corresponding primary key $PK$ in the **referenced relation** $R_2$
- a **null.** The $FK$ in $R_1$ typically should **not** be its own primary key.

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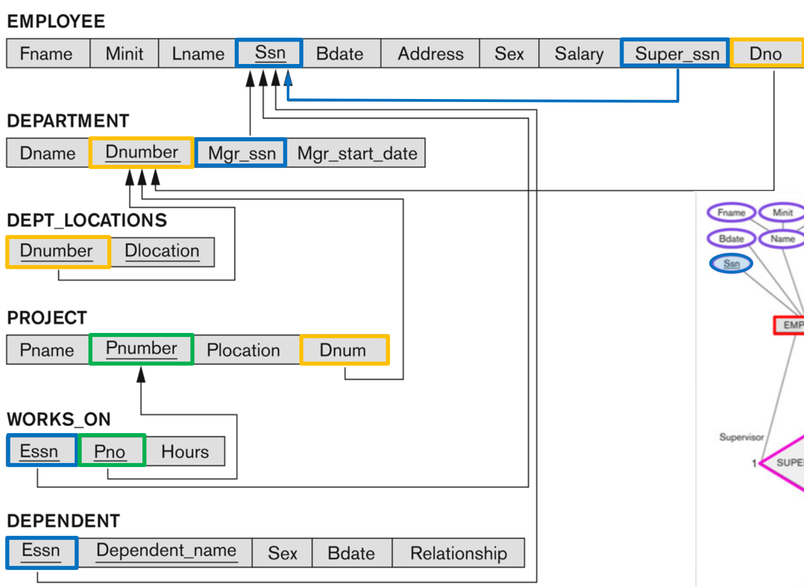
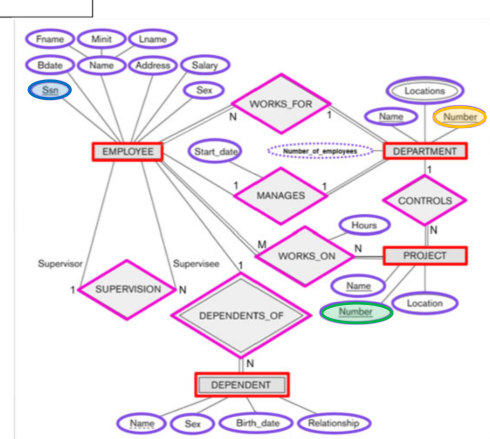
## Referential Integrity Constraints for COMPANY Database

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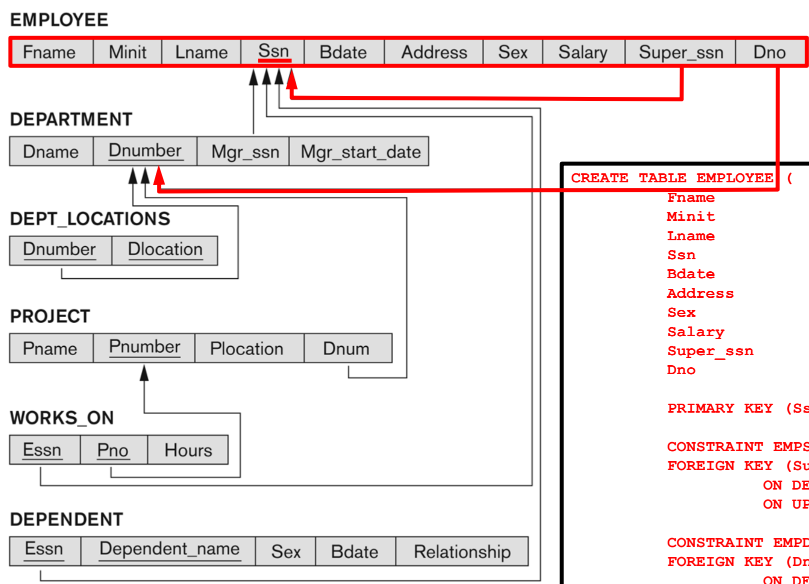


**SQL**

```
CREATE TABLE EMPLOYEE (
          Fname                VARCHAR(15)          NOT NULL,
          Minit                CHAR,
          Lname                VARCHAR(15)          NOT NULL,
          Ssn                  CHAR(9)              NOT NULL,
          Bdate                DATE,
          Address              VARCHAR(30),
          Sex                  CHAR,
          Salary               DECIMAL(10,2),
          Super_ssn            CHAR(9),
          Dno                  INT                  NOT NULL,

          PRIMARY KEY (Ssn),

          CONSTRAINT EMPSUPERFK
          FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE(Ssn)
                    ON DELETE SET NULL
                    ON UPDATE CASCADE,

          CONSTRAINT EMPDEPTFK
          FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dnumber)
                    ON DELETE SET DEFAULT
                    ON UPDATE CASCADE
);
```

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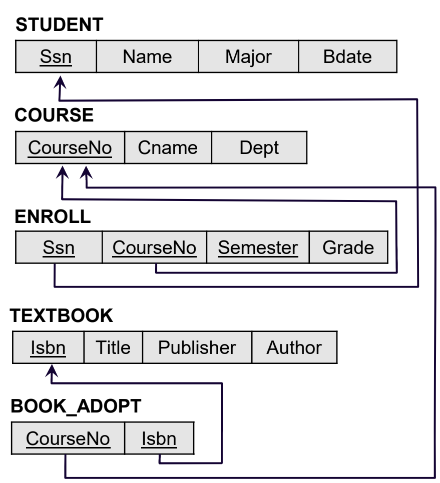
# Example

Relations in a database that keeps track of student enrollment in courses and the books adopted for each course.

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## Update Operations on Relations

- INSERT a tuple.
- DELETE a tuple.
- MODIFY a tuple.
- Integrity constraints should not be violated by the update operations.
- Several update operations may have to be grouped together.
- Updates may **propagate** to cause other updates automatically. This may be necessary to maintain integrity constraints.

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## Update Operations on Relations

In case of integrity violation, several actions can be taken:

- Cancel the operation that causes the violation (RESTRICT or REJECT option)
- Perform the operation but inform the user of the violation
- Trigger additional updates so the violation is corrected (CASCADE option, SET NULL option)
- Execute a user-specified error-correction routine

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NJIT New Jersey Institute of Technology

## INSERT Examples

***Operation***:
Insert <'Cecilia', 'F', 'Kolonsky', NULL, '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, NULL, 4> into EMPLOYEE.
***Result***: Violates the entity integrity constraint (NULL for the primary key Ssn).

***Operation***:
Insert <'Cecilia', 'F', 'Kolonsky', '999887777', '1960-04-05', '6357 Windy Lane, Katy, TX', F, 28000, '987654321', 4> into EMPLOYEE.
***Result***: Violates the key constraint because another tuple with the same Ssn value already exists in the EMPLOYEE relation.

***Operation***:
Insert <'Cecilia', 'F', 'Kolonsky', '677678989', '1960-04-05', '6357 Windswept, Katy, TX', F, 28000, '987654321', 7> into EMPLOYEE.
***Result***: Violates the referential integrity constraint specified on Dno in EMPLOYEE because no corresponding referenced tuple exists in DEPARTMENT with Dnumber = 7.

EMPLOYEE

| Fname | Minit | Lname | Ssn | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|---|---|---|---|---|---|---|---|---|---|
| John | B | Smith | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M | 30000 | 333445555 | 5 |
| Franklin | T | Wong | 333445555 | 1955-12-08 | 638 Voss, Houston, TX | M | 40000 | 888665555 | 5 |
| Alicia | J | Zelaya | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX | F | 25000 | 987654321 | 4 |
| Jennifer | S | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX | F | 43000 | 888665555 | 4 |
| Ramesh | K | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M | 38000 | 333445555 | 5 |
| Joyce | A | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX | F | 25000 | 333445555 | 5 |
| Ahmad | V | Jabbar | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX | M | 25000 | 987654321 | 4 |
| James | E | Borg | 888665555 | 1937-11-10 | 450 Stone, Houston, TX | M | 55000 | NULL | 1 |

DEPT_LOCATIONS

| Dnumber | Dlocation |
|---|---|
| 1 | Houston |
| 4 | Stafford |
| 5 | Bellaire |
| 5 | Sugarland |
| 5 | Houston |

WORKS_ON

| Essn | Pno | Hours |
|---|---|---|
| 123456789 | 1 | 32.5 |
| 123456789 | 2 | 7.5 |
| 666884444 | 3 | 40.0 |
| 453453453 | 1 | 20.0 |
| 453453453 | 2 | 20.0 |
| 333445555 | 2 | 10.0 |
| 333445555 | 3 | 10.0 |
| 333445555 | 10 | 10.0 |
| 333445555 | 20 | 10.0 |
| 999887777 | 30 | 30.0 |
| 999887777 | 10 | 10.0 |
| 987987987 | 10 | 35.0 |
| 987987987 | 30 | 5.0 |
| 987654321 | 30 | 20.0 |
| 987654321 | 20 | 15.0 |
| 888665555 | 20 | NULL |

DEPARTMENT

| Dname | Dnumber | Mgr_ssn | Mgr_start_date |
|---|---|---|---|
| Research | 5 | 333445555 | 1988-05-22 |
| Administration | 4 | 987654321 | 1995-01-01 |
| Headquarters | 1 | 888665555 | 1981-06-19 |

PROJECT

| Pname | Pnumber | Plocation | Dnum |
|---|---|---|---|
| ProductX | 1 | Bellaire | 5 |
| ProductY | 2 | Sugarland | 5 |
| ProductZ | 3 | Houston | 5 |
| Computerization | 10 | Stafford | 4 |
| Reorganization | 20 | Houston | 1 |
| Newbenefits | 30 | Stafford | 4 |

DEPENDENT

| Essn | Dependent_name | Sex | Bdate | Relationship |
|---|---|---|---|---|
| 333445555 | Alice | F | 1986-04-05 | Daughter |
| 333445555 | Theodore | M | 1983-10-25 | Son |
| 333445555 | Joy | F | 1958-05-03 | Spouse |
| 987654321 | Abner | M | 1942-02-28 | Spouse |
| 123456789 | Michael | M | 1988-01-04 | Son |
| 123456789 | Alice | F | 1988-12-30 | Daughter |
| 123456789 | Elizabeth | F | 1967-05-05 | Spouse |



## Possible Violations for INSERT

INSERT may violate any of the constraints:
- Domain constraint:
  - if one of the attribute values provided for the new tuple is not of the specified attribute domain
- Key constraint:
  - if the value of a key attribute in the new tuple already exists in another tuple in the relation
- Entity integrity:
  - if the primary key value is null in the new tuple
- Referential integrity:
  - if a foreign key value in the new tuple uses a primary key value that does not exist in the referenced relation

## DELETE Examples

***Operation***:
Delete the WORKS_ON tuple with Essn = '999887777' and Pno = 10.
***Result***: Acceptable. Deletes exactly one tuple.

***Operation***:
Delete the EMPLOYEE tuple with Ssn = '333445555'.
***Result***: Will result in referential integrity violations, because this Ssn is referenced by tuples from the EMPLOYEE, DEPARTMENT, WORKS_ON and DEPENDENT relations.

***Operation***:
Delete the EMPLOYEE tuple with Ssn = '999887777'.
***Result***: Will result in referential integrity violations, because there are tuples in WORKS_ON that refer to this Ssn.

**EMPLOYEE**

| Fname | Minit | Lname | Ssn | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|---|---|---|---|---|---|---|---|---|---|
| John | B | Smith | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M | 30000 | 333445555 | 5 |
| Franklin | T | Wong | 333445555 | 1955-12-08 | 638 Voss, Houston, TX | M | 40000 | 888665555 | 5 |
| Alicia | J | Zelaya | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX | F | 25000 | 987654321 | 4 |
| Jennifer | S | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX | F | 43000 | 888665555 | 4 |
| Ramesh | K | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M | 38000 | 333445555 | 5 |
| Joyce | A | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX | F | 25000 | 333445555 | 5 |
| Ahmad | V | Jabbar | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX | M | 25000 | 987654321 | 4 |
| James | E | Borg | 888665555 | 1937-11-10 | 450 Stone, Houston, TX | M | 55000 | NULL | 1 |

**DEPT_LOCATIONS**

| Dnumber | Dlocation |
|---|---|
| 1 | Houston |
| 4 | Stafford |
| 5 | Bellaire |
| 5 | Sugarland |
| 5 | Houston |

**WORKS_ON**

| Essn | Pno | Hours |
|---|---|---|
| 123456789 | 1 | 32.5 |
| 123456789 | 2 | 7.5 |
| 666884444 | 3 | 40.0 |
| 453453453 | 1 | 20.0 |
| 453453453 | 2 | 20.0 |
| 333445555 | 2 | 10.0 |
| 333445555 | 3 | 10.0 |
| 333445555 | 10 | 10.0 |
| 333445555 | 20 | 10.0 |
| 999887777 | 30 | 30.0 |
| 999887777 | 10 | 10.0 |
| 987987987 | 10 | 35.0 |
| 987987987 | 30 | 5.0 |
| 987654321 | 30 | 20.0 |
| 987654321 | 20 | 15.0 |
| 888665555 | 20 | NULL |

**DEPARTMENT**

| Dname | Dnumber | Mgr_ssn | Mgr_start_date |
|---|---|---|---|
| Research | 5 | 333445555 | 1988-05-22 |
| Administration | 4 | 987654321 | 1995-01-01 |
| Headquarters | 1 | 888665555 | 1981-06-19 |

**PROJECT**

| Pname | Pnumber | Plocation | Dnum |
|---|---|---|---|
| ProductX | 1 | Bellaire | 5 |
| ProductY | 2 | Sugarland | 5 |
| ProductZ | 3 | Houston | 5 |
| Computerization | 10 | Stafford | 4 |
| Reorganization | 20 | Houston | 1 |
| Newbenefits | 30 | Stafford | 4 |

**DEPENDENT**

| Essn | Dependent_name | Sex | Bdate | Relationship |
|---|---|---|---|---|
| 333445555 | Alice | F | 1986-04-05 | Daughter |
| 333445555 | Theodore | M | 1983-10-25 | Son |
| 333445555 | Joy | F | 1958-05-03 | Spouse |
| 987654321 | Abner | M | 1942-02-28 | Spouse |
| 123456789 | Michael | M | 1988-01-04 | Son |
| 123456789 | Alice | F | 1988-12-30 | Daughter |
| 123456789 | Elizabeth | F | 1967-05-05 | Spouse |

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## Possible Violations for DELETE

DELETE may violate only referential integrity:

- If the primary key value of the tuple being deleted is used by other tuples (foreign keys) in the database
  - Can be remedied by several actions:
    - **RESTRICT**: reject the deletion
    - **CASCADE**: delete all other referencing tuples
    - **SET NULL**: set the foreign keys of the referencing tuples to NULL
- One of the above must be specified during database design for each foreign key constraint

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## MODIFY Examples

***Operation***:
Modify the salary of the EMPLOYEE tuple with Ssn = '999887777' to 28000.
***Result***: Acceptable.

***Operation***:
Modify the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 7.
***Result***: Unacceptable, because it violates referential integrity.

***Operation***:
Modify the Dno of the EMPLOYEE tuple with Ssn = '999887777' to 1.
***Result***: Acceptable.

***Operation***:
Update the Ssn of the EMPLOYEE tuple with Ssn = '999887777' to '987654321'.
***Result***: Unacceptable, because it violates the key constraint by repeating a value that already exists as a primary key in another tuple. It also violates referential integrity constraints because there are other relations that refer to the existing value of Ssn.

**EMPLOYEE**

| Fname | Minit | Lname | Ssn | Bdate | Address | Sex | Salary | Super_ssn | Dno |
|---|---|---|---|---|---|---|---|---|---|
| John | B | Smith | 123456789 | 1965-01-09 | 731 Fondren, Houston, TX | M | 30000 | 333445555 | 5 |
| Franklin | T | Wong | 333445555 | 1955-12-08 | 638 Voss, Houston, TX | M | 40000 | 888665555 | 5 |
| Alicia | J | Zelaya | 999887777 | 1968-01-19 | 3321 Castle, Spring, TX | F | 25000 | 987654321 | 4 |
| Jennifer | S | Wallace | 987654321 | 1941-06-20 | 291 Berry, Bellaire, TX | F | 43000 | 888665555 | 4 |
| Ramesh | K | Narayan | 666884444 | 1962-09-15 | 975 Fire Oak, Humble, TX | M | 38000 | 333445555 | 5 |
| Joyce | A | English | 453453453 | 1972-07-31 | 5631 Rice, Houston, TX | F | 25000 | 333445555 | 5 |
| Ahmad | V | Jabbar | 987987987 | 1969-03-29 | 980 Dallas, Houston, TX | M | 25000 | 987654321 | 4 |
| James | E | Borg | 888665555 | 1937-11-10 | 450 Stone, Houston, TX | M | 55000 | NULL | 1 |

**DEPT_LOCATIONS**

| Dnumber | Dlocation |
|---|---|
| 1 | Houston |
| 4 | Stafford |
| 5 | Bellaire |
| 5 | Sugarland |
| 5 | Houston |

**WORKS_ON**

| Essn | Pno | Hours |
|---|---|---|
| 123456789 | 1 | 32.5 |
| 123456789 | 2 | 7.5 |
| 666884444 | 3 | 40.0 |
| 453453453 | 1 | 20.0 |
| 453453453 | 2 | 20.0 |
| 333445555 | 2 | 10.0 |
| 333445555 | 3 | 10.0 |
| 333445555 | 10 | 10.0 |
| 333445555 | 20 | 10.0 |
| 999887777 | 30 | 30.0 |
| 999887777 | 10 | 10.0 |
| 987987987 | 10 | 35.0 |
| 987987987 | 30 | 5.0 |
| 987654321 | 30 | 20.0 |
| 987654321 | 20 | 15.0 |
| 888665555 | 20 | NULL |

**DEPARTMENT**

| Dname | Dnumber | Mgr_ssn | Mgr_start_date |
|---|---|---|---|
| Research | 5 | 333445555 | 1988-05-22 |
| Administration | 4 | 987654321 | 1995-01-01 |
| Headquarters | 1 | 888665555 | 1981-06-19 |

**PROJECT**

| Pname | Pnumber | Plocation | Dnum |
|---|---|---|---|
| ProductX | 1 | Bellaire | 5 |
| ProductY | 2 | Sugarland | 5 |
| ProductZ | 3 | Houston | 5 |
| Computerization | 10 | Stafford | 4 |
| Reorganization | 20 | Houston | 1 |
| Newbenefits | 30 | Stafford | 4 |

**DEPENDENT**

| Essn | Dependent_name | Sex | Bdate | Relationship |
|---|---|---|---|---|
| 333445555 | Alice | F | 1986-04-05 | Daughter |
| 333445555 | Theodore | M | 1983-10-25 | Son |
| 333445555 | Joy | F | 1958-05-03 | Spouse |
| 987654321 | Abner | M | 1942-02-28 | Spouse |
| 123456789 | Michael | M | 1988-01-04 | Son |
| 123456789 | Alice | F | 1988-12-30 | Daughter |
| 123456789 | Elizabeth | F | 1967-05-05 | Spouse |

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## Possible Violations for MODIFY

MODIFY may violate the **domain constraint** and NOT NULL constraint on an attribute

Any of the other constraints may also be violated, depending on the attribute being modified:

- Modifying the primary key (PK):
  - Similar to a DELETE followed by an INSERT
  - Need to specify similar options to DELETE
- Modifying a foreign key (FK):
  - May violate referential integrity
- Modifying an ordinary attribute (neither PK nor FK):
  - Can only violate domain constraints

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```
CREATE TABLE EMPLOYEE (
        Fname           VARCHAR(15)     NOT NULL,
        Minit           CHAR,
        Lname           VARCHAR(15)     NOT NULL,
        Ssn             CHAR(9)         NOT NULL,
        Bdate           DATE,
        Address         VARCHAR(30),
        Sex             CHAR,
        Salary          DECIMAL(10,2),
        Super_ssn       CHAR(9),
        Dno             INT             NOT NULL,

        PRIMARY KEY (Ssn),

        CONSTRAINT EMPSUPERFK
        FOREIGN KEY (Super_ssn) REFERENCES EMPLOYEE(Ssn)
                ON DELETE SET NULL
                ON UPDATE CASCADE,

        CONSTRAINT EMPDEPTFK
        FOREIGN KEY (Dno) REFERENCES DEPARTMENT(Dnumber)
                ON DELETE SET DEFAULT
                ON UPDATE CASCADE
);
```

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